SYMBIOSIS UNIVERSITY OF APPLIED SCIENCES

INDORE

PROJECT REPORT

ON

OPTICAL HEART RATE DETECTION using WEBCAM

Submitted to "Symbiosis University of Applied Sciences, Indore
As an Internship report for the partial fulfillment of the award of degree of

BACHELOR OF TECHNOLOGY

IN

SCHOOL OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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CERTIFICATE

This is to certify that the Internship report entitled OPTICAL HEART RATE DETECTION using WEBCAM, submitted by Abhishek Patwa , student of fourth year towards partial fulfillment of the degree of Bachelor of Technology in School of Computer Science and Information Technology in year 2018-2022 Symbiosis University of Applied Sciences , Indore (M.P.) is in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology and is a bonafide record of the work carried by Abhishek Patwa, Neelam Kukreja and Garvita Jain during the academic semester eighth.

Place: Indore

Date: 14/5/2021

INTERNAL EXAMINER EXTERNAL EXAMINER

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RECOMMENDATION

The work entitled "OPTICAL HEART RATE DETECTION using WEBCAM", submitted by Abhishek Patwa, Neelam Kukreja and Garvita Jain, students of fourth year Computer Science and Information Technology, towards the partial fulfillment for the award of degree of Bachelor of Technology in Computer Science and Information Technology of Symbiosis University of Applied Sciences Indore(M.P.) is a satisfactory account of their Internship and is recommended for the award of the degree.

Endorsed By:

Dr. Sujatha R Upadhyaya

Director, SCSIT

Student Undertaking

I hereby undertake that the project work entitled "OPTICAL HEART RATE DETECTION using WEBCAM" has been carried out by me from the period15/2/2021 to14/5/2021 and the report so prepared is a record of work done by me during my internship. I further declare that I have completed the internship in accordance with the Internship policy of the University. This Project report is submitted towards fulfillment of my academic requirement and not for any other purpose.

I hereby undertake that the material of this Project is my original work and I have not copied anything from anywhere else. The material obtained from other sources has been duly acknowledged. I understand that if at any stage, it is found that I have indulged in any malpractice or the project and the project report has been copied or not completed by me, the university shall cancel my degree/withhold my result and appropriate disciplinary action shall be initiated against me.

Student Name and Signature **Mentor Name & Signature**

Enrollment Number:

Name of School : CSIT

Date: 14/5/21

Head of School Name & Signature

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Positive inspiration and right guidance are must in every aspect of life. Especially, when we arrive at academic stage for instance. For the success of our project a number of obligations have been taken. We have performed solemn duty of expressing a heartfelt thanks to all who have endowed us with their precious perpetual guidance, suggestions and information. Any kind of help directly or indirectly has proved importance to us.

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Chapter 1: Introduction

1.1 INTRODUCTION

Now-a-days health problems like cardiac failure, lung failures & heart related diseases are arising day by day at a very high rate. Due to these problems time to time health monitoring is very essential. A modern concept is health monitoring of a patient wirelessly. It is a major development in medical arena. Health professionals have developed a brilliant and inexpensive health monitoring system or providing more comfortable living to the people suffering from various diseases using leading technologies like wireless communications, wearable and portable remote health monitoring device. As visits of doctors to the patients constantly are decreased as the information regarding patient's health directly reaches to doctor's monitor screen from anywhere the patient resides. Also, based on this doctors can save many lives by imparting them a quick & valuable service.

According to the recent statistics, nearly two million people suffer from heart attack every year and one person dies every 33 seconds in India. World Health Organization (WHO) reports that heart disease rate might increase to 23.3% worldwide by the year 2030. The treatment of such chronic disease requires continuous and long term monitoring to have proper control on it. IoT helps to move from manual heart rate monitoring systems to remote heart rate monitoring systems A doctor may not be present all the time to provide medication or treatment to the patients or a guardian may not be present all the time to take the patient to the hospital. Hence, our proposed system is the right solution for this problem.

In today's era, health problems are increasing day-by-day at a high pace. The death rate of 55.3 million people dying each year or 151,600 people dying each day or 6316 people dying each hour is a big issue for all over the world. Hence it is the need of hour to overcome such problems. This paper presents a real time HR monitoring method using a webcam of a laptop computer. The heart rate is obtained through facial skin color variation caused by blood circulation. Three different signal processing methods such as Fast Fourier Transform (FFT), Independent Component Analysis (ICA) and Principal Component Analysis (PCA) have been applied on the color channels in video recordings and the blood volume pulse (BVP) is extracted from the facial regions. Heart rate is subsequently quantified and compared to corresponding reference measurements.

The obtained results show that there is a high degree of agreement between the proposed experiments and reference measurements

1.2 Problem Definition

- People across the world are currently dealing with an unprecedented rise in heart related diseases and require methods to combat the situation.
- The amount of time and resources required to perform simple operations such as heart rate measurement are deemed to be in excess from a sociological point of view.
- People require a system that allows them to measure their heart rates remotely and quickly as having to visit a nearby clinic or hospital comes with its own hurdles and in may lead to unnecessary complications.

Chapter 2: The Project

2.1 Project Definition

Optical Heart Rate Detection Using WEBCAM

2.1.1 Objective

- Developing a real time HR monitoring method using a webcam of a laptop computer where the heart rate shall be obtained through facial skin color variation caused by blood circulation.
- Making sure the system is easy to use to ensure access to people of all ages without facing any issues with its operation.
- Making sure that the final product helps us save both time and resources on a large scale in comparison to the current contact based heart rate measuring systems, making its application viable on a national scale.

2.1.2 Project Scope

Cardiac pulse is typically measured in clinical settings using electrocardiogram (ECG), which requires patients to wear chest straps with adhesive gel patches that can be abrasive and become uncomfortable for the user. Heart rate may also be monitored using pulse oximetry sensors that may be worn on the fingertip or earlobe. These sensors are not convenient for long-term wear and the pressure can become uncomfortable over time. In addition to the discomforts of traditional pulse measurement devices, these devices can damage the fragile skin of premature newborns or elderly people.

For these people especially, a non-contact means of detecting pulse could be very beneficial. Non-contact heart rate measurement through a simple webcam or phone camera would also aid telemedicine and allow the average person to track their heart rate without purchasing special equipment. As part of the recent gain in popularity of fitness apps and the quantified self, regular non-obtrusive monitoring through a computer or phone camera may help detect changes in a person's heart rate over time and indicate changing health or fitness.

Heart rate can be detected without contact through photo-plethysmography (PPG), which measures variations in blood volume by detecting changes in light reflectance or transmission throughout the cardiovascular pulse cycle.

PPG is usually performed with dedicated light sources with red or infrared wavelengths, as is the case for pulse oximetry sensors. Plethysmographic signal can also be detected in video from a regular color camera. The signal can be detected within the red, green, and blue channels of color video of exposed skin, but that is strongest in the green channel, which corresponds to the fact that hemoglobin has absorption peaks for green and yellow light wavelengths. Although the signal can be detected in multiple locations on the body, it is strongest on the face, especially the forehead. Although the plethysmographic signal may be detected in the raw color channel data, it is mixed in with other sources of color variation such as changes in ambient light or motion. The signal can be better extracted by using independent component analysis (ICA) to separate independent source signals from the mixed color signals.

2.1.3 Project Algorithm

(1) Perform face detection on one frame of the video

(2) Decompose this frame into sub frames using the Gaussian pyramid decomposition algorithm. Thus the highest level of the pyramid has low pass information.

(3) Filter each level of the pyramid using a Band Pass Filter, with cut-off frequency given as [(hifreq*frame rate/sampling rate)±1] .

(4) Magnify the intensities of the frequency band of interest by a magnification factor.

(5) Collapse the pyramid into a single frame.

(6) Repeat the above 5 steps for all the remaining frames of the video.

Chapter 3: Requirements Analysis

3.1 Hardware Requirements

A webcam or a laptop's built-in camera is the only hardware requirement.

1.2 Software Requirements

- The software is implemented on a Windows environment using Python programming language aided by three libraries: OpenCV, Numpy and Scipy.
- The OpenCV library provides the necessary resources to the facial detection algorithm and the frame capture through the webcam connected to a computer. Numpy and Scipy are used for optimization, stats and signal processing.
- Although it was not tested here, the software developed does not rely on any OS specific functionality, and thus can be ported to a Linux environment.

3.3 Non Functional Requirements

#	Type of Requirement	Details
1	Performance Requirements	*Since the software works in real-time, the capture can take about a minute. This time is dependent on the system's processing capabilities.*
2	Scalability Requirements	*A single person should be in the frame for capture. Incase multiple people are in the frame the software shall automatically focus on a single one. The software has not been scaled.*
3	Security Requirements	*The software does not require any user identity authentication or Network IP requirements since no data is collected or stored.*

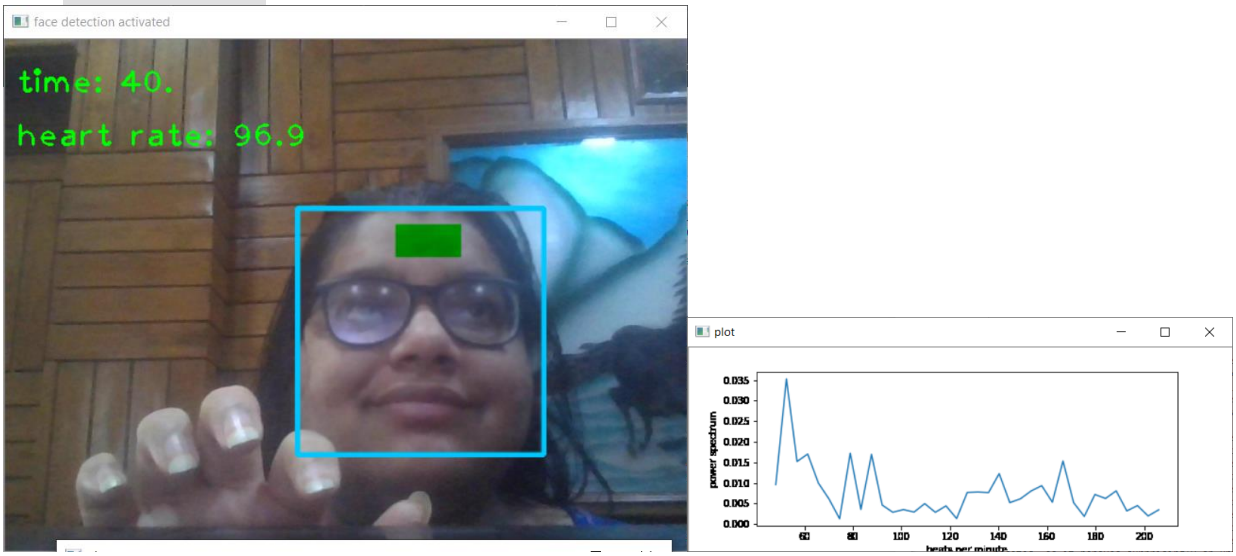
Chapter 4: Design & Features

The main features of the proposed method are 3 independent signals which are called Red signals, Green signals and Blue signals and these signals were produced from the red, blue and green color values of each pixel of all the facial image frames.

#	Title of System Feature	Description	Priority	Functional Requirements
1	*Frame capture*	*The first stage consists of the acquisition of a frame through a webcam connected to the computer. As it is usual with most webcams, the captured frame is in the RGB color format*	*High*	*Although the resolution of webcam is not so important (a webcam with a 1280 × 720 resolution and 30 fps is used in the experiments), it is paramount that the webcam is able to capture the raw data, without any compression, as this would result in the loss of the color variations we are seeking.* *This is usually not a problem for webcams that are directly connected to the computer, but it must be observed for webcams connected through any kind of wired or wireless networks.*
2	*Face Detection and ROI Definition*	*The second stage consists of the facial detection over the acquired frame. As default, only one face is to be detected in this project, ignoring any other subjects appearing in the image. The third stage consists of defining a bounding box over the previously detected face.*	*Medium*	*The ROI is defined as h × w, where h is the height of the bounding box of the detected face and w is 55% of the width of this bounding box. The position of the ROI is centralized within the detected face* *The face detection algorithm is only performed for the first frame in each second (i.e., once every 30 frames). The ROI is kept the same for the remaining frames.*
3	*Trace Extraction*	*In the fourth stage, for every frame n, the ROI consists of a color image h×w with three channels, Red, Green and Blue.*	*High*	*The image is normalized, removing the mean value for each channel, making the variance between samples unitary as follows:* $Cbi(n) = Ci(n) - \mu i \sigma i$ *where i is for R, G and B, μi and σi are the mean and standard deviation for each channel, respectively.*
4	*Frequency Domain Filtering:*	*In the fifth stage, the signal of interest is the trace, T (n). These algorithms are not applied for every new trace sample (i.e., every frame), but rather once every two seconds (i.e., once every 60 frames).*	*High*	*The algorithm takes the trace values for the last 10 seconds and store it in a vector. It is then transformed to the frequency domain with a Fast Fourier Transform - FFT of 2048 points.*
5	*Peak Detection and Recurring Point of Interest Analysis*	*This is the sixth stage, which performs the heart rate estimation by analyzing the existing peaks in the obtained signal after the previous stage.*	*High*	*It is triggered only when the FFT is performed (i.e., every 2 seconds). The analyzed range is between 31 and 185 bpm, which corresponds to [0.52 3.10] Hz. The five magnitude peaks have their X-axis values (the bpm) stored and are considered as points of interest*

Chapter 5: Experiment and Testing

5.1 Test Case



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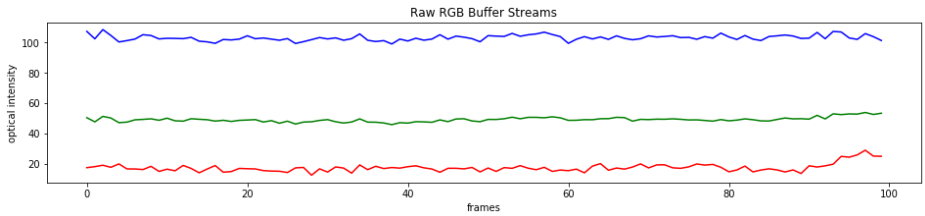

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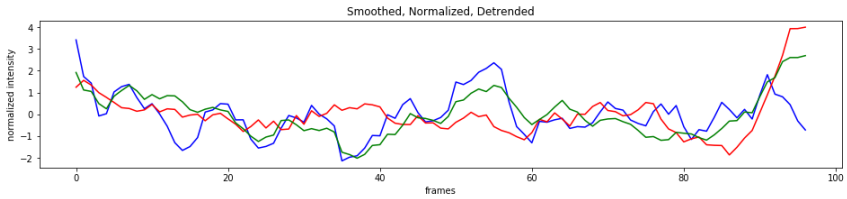

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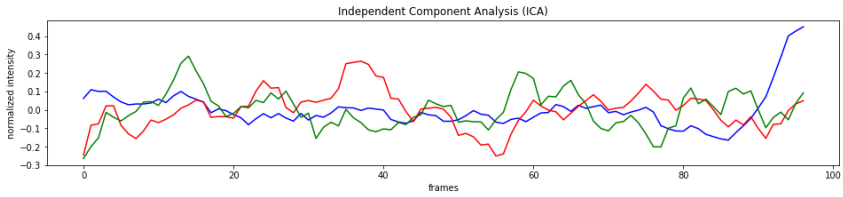

Chapter 6: Conclusion

6.1 Problems and Issues

PPG is known to be susceptive to motion-induced signal corruption and overcoming motion artifacts presents one of the most challenging problems. A good Signal-to-Noise Ratio (SNR) is essential to accurately detect the heart rate signal. It can be improved by either reducing the noise or increasing the signal power. In our case, the noise affecting the measurement comes from three sources basically:

- Image noise: It originates in the camera sensor. By averaging all the pixels for the brightness calculation, we are filtering out a big part of its spatial component. The band-pass filter that we are applying to the brightness signal is also rejecting much of its time component. However, there will always be some level of this noise in the band of interest. User behavior:
- Pressure variation: The pressure variations of the fingertip against the camera lens may show up in the signal. If the user has shaky hands, it is better that she uses a finger of the hand holding the phone.
- Lighting variation: Any change in the light sources or in the scene reflecting that light towards the camera may introduce noise in our pass band. For instance, moving the phone in the air while recording the video sequence may introduce artifacts if the scene parts in the field of view have very different light intensities.

6.2 Future Scope

- The software shall be further updated to assess multiple users simultaneously and display their respective heart rates without conflict.
- Additional parameters shall be defined in order to invalidate the readings in cases where the user exits the frame while the process is still underway.
- Given the application of optimal equipment the image noise can be reduced to a negligible amount granting greater accuracy in readings.

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